



QUALITY SYSTEM

FORM

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TITLE:

LABORATORY OOS INVESTIGATION FORM

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OOS Number: OOS- 262074

Phase I

Section A – Test Details

Initiator:	Alex Saravia (Written by: Qiyue Chen)	Test Date:	31-Aug-2026	Date Initiated:	07-Sep-2026
Date of Incident:	07-Sep-2026	Test Name:	USP <71> / EP 2.6.1 Sterility Test	Sample ID:	ETX-260806-0702 (BUD-T30)
Name of Analyst who Performed the Test:	Prepping Analyst: Andrew Carrillo (AC)	Sample / Active Name	TB-500 10mg/ml	Lot #:	24JUL2026-059
	Processing Analyst: Alex Saravia (ES)	Dosage Form	Liquid		
Description of Incident:	On 07 Sep 2026, sample ETX-260806-0702 (BUD-T30) was found positive for viable microorganisms after USP <71> sterility testing.	☒ Client Care notified of OOS			
		☒ Sterility and Endotoxin test related OOS; Client must be notified within 24 hours of OOS			
	SOP / Test Method # MICRO-SOP-5	Effective Date: 24Jul2026	SOP / Test Method Rev: 19	Limits/Specification: Pass/Fail	
Manager notified	Manager/designee name: Kathan Parikh	Date: 07-Sep-2026	Quality Assurance Department notified:	Yes ☒ No ☐	

Section B – Investigation and Root Cause Analysis

Personnel Discrepancy		Comments	
Analyst interviewed?	☒ Yes ☐ No ☐ N/A	Yes, analysts Andrew Carrillo and Alex Saravia were interviewed comprehensively.	
Correct samples analyzed?	☒ Yes ☐ No ☐ N/A	Yes, sample ID: ETX-260806-0702	
Correct SOP / Test Method used/followed?	☒ Yes ☐ No ☐ N/A	Yes, as per MICRO-SOP-5	
Correct sample collection/technique?	☒ Yes ☐ No ☐ N/A	Yes, as per MICRO-SOP-5	
Are the Test Result Records Attached & Calculations Verified and Correct?	☒ Yes ☐ No ☐ N/A	Yes, Information is available in EagleTrax under ETX-260806-0702	
Is the analyst qualified to perform the test?	☒ Yes ☐ No ☐ N/A	Yes, the analysts are trained and qualified by quality to perform the test	
Is formula worksheet provided and verified?	☐ Yes ☐ No ☒ N/A	Not Applicable	
Does the processing method/method set conform to the required specification?	☐ Yes ☐ No ☒ N/A	Not Applicable	
Was the sample stored appropriately?	☒ Yes ☐ No ☐ N/A	Yes, the sample was stored as per client's instructions. Information is available in EagleTrax Sample Location History under ETX-260806-0702	
Material Discrepancy		Lot Number	Expiry Date
Is the integrity of the sample(s) questionable?	☐ Yes ☒ No ☐ N/A		
Was the sample transported appropriately?	☒ Yes ☐ No ☐ N/A		
Are the other test results from this run under investigation?	☐ Yes ☒ No ☐ N/A		



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Appropriate/correct glassware used?	☐ Yes ☐ No ☒ N/A		
Are reagents stored in proper condition and not expired?	☒ Yes ☐ No ☐ N/A	Process Consumables: Information is available in EagleTrax Sample Consumables section Environmental Plates: TSA and Surface Plate: see attached environmental logs Monthly Cleaning: H2O2 strips and IPA: See attached monthly cleaning logs	Process Consumables: Information is available in EagleTrax Sample Consumables section Environmental Plates: TSA and Surface Plate: see attached environmental logs Monthly Cleaning: H2O2 strips and IPA: See attached monthly cleaning logs
Standard Used	Not applicable	Not applicable	Not applicable
Solvent Used	Not applicable	Not applicable	Not applicable
Is the standard the source of the root cause?	☐ Yes ☐ No ☒ N/A		
Has the analyst been instructed to keep the original test solution for possible retesting?	☐ Yes ☐ No ☒ N/A		
Equipment Discrepancy		Equipment ID	Calibration Due Date
Was the instrument set up correctly?	☒ Yes ☐ No ☐ N/A	ISO 5 BSC E001314	Jun 2027
Is the clean room facility certified?	☒ Yes ☐ No ☐ N/A	CR E001737	Dec 2026
System Performance Check (if applicable)?	☐ Yes ☐ No ☒ N/A	Not applicable	Not applicable
System Suitability RSD < 2% (PDA)?	☐ Yes ☐ No ☒ N/A	Not applicable	Not applicable
System Suitability RSD < 10% (QDA, CAD)?	☐ Yes ☐ No ☒ N/A	See Phase I Summary	See Phase I Summary
Do the environmental monitoring results conform to required standards?	☒ Yes ☐ No ☒ N/A	See Phase I Summary	See Phase I Summary
Do the controls (i.e., positive, negative, etc.) conform to the required standards?	☒ Yes ☐ No ☒ N/A	See Phase I Summary	See Phase I Summary
Is equipment/instrument functional and calibrated/qualified/certified/PM?	☒ Yes ☐ No ☐ N/A	Incubator E001356 (Monitored by Sensor E001450) Incubator E001357 (Monitored by Sensor E001449) Incubator E001034 (Monitored by Sensor E001501) Incubator E001031 (Monitored by Sensor E001505)	Jan 2027 Feb 2027 Jan 2027 Feb 2027 Aug 2027 Feb 2027 Aug 2027 Feb 2027

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Are chromatogram/spectra/raw data correct?				☐ Yes ☐ No ☒ N/A	Not applicable	Not applicable
Section C – Evaluation of Laboratory Testing/Hypothesis Testing (if applicable)						
☒ N/A		☐ Re-vialing of original working sample solution				
☐ Re-injection of sample vial		☐ Re-dilution of a working sample solution from the original stock				
☐ Re-injection of original standard vial		☐ Re-dilution of a working standard solution from the original stock standard				
☐ Re-vialing of original working standard solution		☐ Other: N/A QC 18Sep26				
Section D – Phase I Summary						
Phase I Summary: All analysts involved in the prepping, processing, and reading of the sample – Andrew Carrillo, Alex Saravia and Andrew Carrillo, were interviewed comprehensively. Their answers are recorded throughout this document. Upon arrival, the sample was stored in accordance with the Client ’ s instructions. Analysts - Andrew Carrillo and Alex Saravia - verified the integrity of the sample throughout both the preparation and processing stages. No leaks or turbidity were observed at any point, verifying the integrity of the sample. All reagents and supplies mentioned in the material section above were stored according to the suppliers ’ recommendations, and their integrity was visually verified before utilization. Moreover, all reagents and supplies had valid expiration dates. The functionality of all equipment was confirmed by reviewing data generated by our comprehensive in-house continuous monitoring system. The cleanroom suite (CR 115) used for processing procedures comprises three interconnected sections: the innermost ISO 7 cleanroom (115B), which connects to the middle ISO 7 buffer room (115A), and then to the outermost ISO 8 anteroom (115). A positive air pressure system is maintained throughout the suite to ensure controlled, unidirectional airflow from 115B through 115A and into 115. The ISO 5 BSC E001314, located in the innermost ISO 7 room, (115B), was used for sample processing steps. It was thoroughly cleaned and disinfected prior to each procedure in accordance with MICRO SOP-9 (Cleaning and Disinfecting Procedure for Microbiology). Additionally, BSC E001314 was certified and approved by both the Engineering and Quality Assurance teams. Sample processing was conducted in the ISO 5 BSC E001314 in the innermost ISO 7 room, (115B) by Alex Saravia on 31 Aug 2026 as per MICRO-SOP-5 (USP <71> / EP 2.6.1 Sterility Test).						

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Phase I Summary (continued):

On 28 Aug 2026, the sample vials for ETX-260806-0702 were received from the Sample Submissions team and brought into the Sterile Microbiology lab. Upon arrival, each sample vial was sprayed with an acidified bleach disinfectant, placed into pre-disinfected bins, and allowed a 10-minute contact time.

The secondary disinfection happened in the ISO 8 anteroom (115), where the vials were again treated with acidified bleach and provided a 10-minute contact time before processing. Subsequently, the vials were moved into innermost ISO 7 cleanroom (115B). Inside this cleanroom, the processing analyst, Alex Saravia, performed a final disinfection step, allowing an additional 10-minute contact time. Once fully disinfected, the vials were transferred into the ISO 5 BSC E001314.

Inside the BSC, each container was aseptically opened, ensuring minimal exposure to the environment. The contents of 10 vials (20 mL per media) were dispensed directly from the container into sterile canisters placed on a vacuum manifold for membrane filtration under method suitability ETX-260623-0160. The sample was processed and tested as per MICRO-SOP-5 (USP <71> / EP 2.6.1 Sterility Test). Media bottles were transferred into incubators E001356 and E001357 to begin incubation.

On 07 Sep 2026 (Day 7 of incubation), microbial growth was observed in one TSB (Tryptic Soy Broth) media bottle by reading analyst, Andrew Carrillo, and confirmed by Microbiology Lab Supervisor, Robin Seymour. The positive TSB bottle for ETX-260806-0702 was submitted for Differential Staining and Microbial Identification under ETX-260907-0017. Differential staining indicated the presence of Gram-positive rods. Definitive identification identified the isolate as *Paenibacillus lautus* Gram (+) rods.

The culture media utilized were within their expiry period. Negative control media bottles for the membrane filtration method were handled, processed, and incubated in the same manner as the test samples. No microbial growth was observed in the corresponding negative controls.

Upon analyzing the environmental monitoring results, no microbial growth was detected on personnel monitoring plates, settling plates, or ISO 5 BSC surface sampling plates on the date of testing (31 Aug 2026), the preceding testing date (28 Aug 2026), or the subsequent testing date (01 Sep 2026). Additionally, no microbial growth was recovered from the weekly surface sampling plate collected on 04 Sep 2026 for Cleanroom 115 (E001737). However, weekly active air monitoring of ISO 8 Room 114 (CR 114) recovered 1 CFU (ETX-260914-0487, pending identification).

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Phase I Summary (continued):

It is important to note that all sample processing activities were performed within the validated ISO 5 BSC E001314 located in the innermost ISO 7 cleanroom (Suite 115B). The test samples do not come into contact with ambient ISO 8 air, as samples are transferred in disinfected, lidded containers through the layered cleanroom suites. Furthermore, the colony recovery occurred in the lower-classified ISO 8 anteroom environment, which is physically segregated from the ISO 5 processing zone.

In addition, considering the facility's layered disinfection program, controlled unidirectional airflow, and cleanroom design, it is unlikely that the microbial contaminant detected in the ISO 8 environment impacted on the sample processed within the ISO 5 area of the ISO 7 cleanroom (Suite 115B).

The absence of contamination on analyst glove plates and work-surface environmental monitoring plates collected on the date of testing indicates that there was no viable transfer of contaminants to the ISO 5 biological safety cabinet (BSC). Additionally, the lack of contamination in other samples processed during the same testing session further demonstrates that the testing environment was operating in optimal condition.

Therefore, considering that no viable organisms were recovered from the ISO 5 environment, personnel monitoring results were acceptable on the date of testing, and the environmental isolates were not recovered from locations or surfaces directly involved in sample preparation or testing, the environmental isolate is not considered a credible source of the contamination detected in the test sample.

The analysts confirmed full compliance with cleaning procedures as outlined in MICRO SOP-9 (Cleaning and Disinfecting Procedure for Microbiology) and MICRO-SOP-5 (USP <71> / EP 2.6.1 Sterility Test).

Monthly cleaning and disinfection of the outermost ISO 8 Anteroom, the middle ISO 7 Buffer room, the innermost ISO 7 cleanroom, and its containing ISO 5 BSCs for Cleanroom suite CR115 was performed on 30 Aug 2026, as per MICRO-SOP-9 (Cleaning and Disinfecting Procedure for Microbiology). During both cleaning cycles, it was documented that all H2O2 indicators passed. This confirms the efficient monthly cleaning of all three parts of Cleanroom suite CR115.

Analyzing a 6-month sample history for Hillstone Pharmacy (E17047), this specific analyte "TB-500 10mg/ml" has had no prior failures using USP <71> / EP 2.6.1 Sterility Test during this period. Crucially, this sample was submitted as part of a Beyond-Use Date (BUD) stability study for lot 24JUL2026-059. The baseline T0 time point (tested on 06 Aug 2026 by analyst G. Li) yielded valid passing results for USP <71> sterility, along with passing USP <1207> Container Closure Integrity and USP <85> Bacterial Endotoxin testing. Furthermore, at the current T30 time point, all other concurrent testing—including Appearance, pH, Potency (TB-500 & Benzyl Alcohol), USP <788> Particulate Matter, and USP <85> Bacterial Endotoxin—met all acceptance criteria and passed. The isolation of microbial growth (*Paenibacillus lautus*) exclusively at the 30-day stability storage interval indicates a true product stability/Beyond-Use Date limitation rather than laboratory contamination during sample manipulation on 31 Aug 2026.

To assess the potential for sample-to-sample contamination contributing to the positive results, a comprehensive review was conducted of all samples processed on the same day (31 Aug 2026). Sample ETX-260806-0702 was the 4th sample processed in the batch. All other samples processed by the analyst and other analysts that day tested negative. These findings suggest that cross-contamination between samples is highly unlikely.

Based on the observations outlined above, it is unlikely that the failing results were due to reagents, supplies, the cleanroom environment, the process, or analyst involvement. Consequently, the possibility of laboratory error contributing to this failure is minimal and the original result is deemed to be valid.



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Has root cause been identified as a possible laboratory error?

Note: Root cause must be conclusive, verifiable, and well-documented.

☐ Yes☐ If retesting is **required**, complete Phase II of the OOS (initiate Form 3.100.019.F02).☒ No, the OOS result is valid.

Reviewer comment(s):

Prepared by (Print):

Alex Saravia (Written by: Qiyue Chen)

Signature/Date:

Lab Manager/Designee (Print):

Robin Seymour

Signature/Date:

Section E – Quality Disposition

Can investigation be closed?

☐ Yes☐ Original result is valid.☐ No, initiate Form 3.100.019.F02.

Approved by QA Representative (Print):

Signature/Date:

Table 1: Information for ETX-260806-0702 (BUD-T30) under investigation

Processing Analyst	Reading Analyst	Sample ID	Related Microbial ID	Media with Microbial growth	Morphology Description
Alex Saravia	Andrew Carrillo	ETX-260806-0702 (BUD-T30)	ETX-260907-0017	1 x 100 mL TSB	Paenibacillus lautus (Gram (+) rods)

Table 2: Environmental Monitoring for Analyst & Cleanroom

Environmental Monitoring (EM) Sampling Site	Frequency	Date (DDMM YYYY)	Analyst (Initials)	Day /Week(s)	Observation	Environmental Monitoring Plate ETX ID	Microbial ID	Notes
Personnel EM Bracketing 31Aug2026								
Personal (Left Touch and Right Touch)	Daily	28 Aug 2026	ES	Date Before Testing	No growth	N/A	N/A	None
Personal (Left Touch and Right Touch)	Daily	31 Aug 2026	ES	Date of Testing	No growth	N/A	N/A	None
Personal (Left Touch and Right Touch)	Daily	01 Sep 2026	ES	Date After Testing	No growth	N/A	N/A	None
Biological Safety Cabinet EM Bracketing Biological Safety Cabinet (BSC) E001314								
Surface Sampling of ISO 5 (4 locations)	Daily	28 Aug 2026	ES	Date Before Testing	No growth	N/A	N/A	None
Surface Sampling of ISO 5 (4 locations)	Daily	31 Aug 2026	ES	Date of Testing	No growth	N/A	N/A	None
Surface Sampling of ISO 5 (4 locations)	Daily	01 Sep 2026	ES	Date After Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	28 Aug 2026	ES	Date Before Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	31 Aug 2026	ES	Date of Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	01 Sep 2026	ES	Date After Testing	No growth	N/A	N/A	None
Weekly Active Air Sampling of CR 115 (E001737)								
Active Air Sampling of Cleanrooms	Weekly	04 Sep 2026	SMO	Week of Testing	1 CFU (ISO 8 114)	ETX-260914-0487	Pending	None
Surface Sampling of Anteroom and Cleanroom of CR 115 (E001737)								
Surface Sampling of Cleanrooms	Weekly	04 Sep 2026	SMO	Week of Testing	No growth	N/A	N/A	None